HW3

Sam Olson

Q1

For unit i, $i = 1, \ldots, n$, let y_i and x_i denote respectively the dependent variable and the known value of an independent variable. For a known value t, let z_i be defined as

$$z_i = \begin{cases} 0 & \text{if } x_i \leq t, \\ 1 & \text{otherwise,} \end{cases}$$

for $i = 1, \ldots, n$. Assume that the y_i's are uncorrelated with common variance σ^2, an unknown positive constant. Consider the (unconstrained) linear model

$$\mathrm{E}(y_i) = \beta_0 + \beta_1 z_i + \beta_2 x_i + \beta_3 x_i z_i, \tag{1}$$

and the constrained linear model

$$\mathrm{E}(y_i) = \gamma_0 + \gamma_1 z_i + \gamma_2 x_i + \gamma_3 x_i z_i, \quad \text{with the constraint} \quad \gamma_1 + \gamma_3 t = 0. \tag{2}$$

You can assume that all β's and γ's are estimable with the available data.

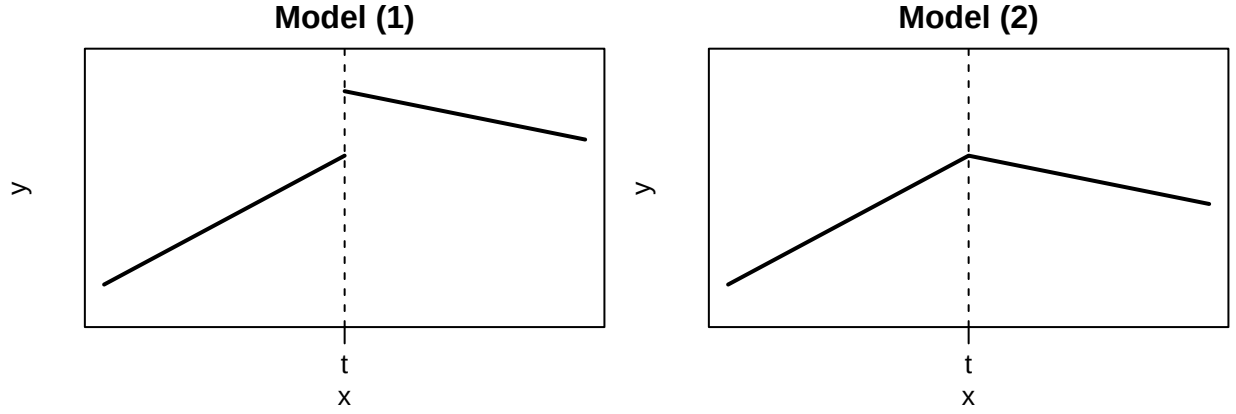
(a)

For each of the two models, sketch in as much detail as possible what relationship the model proposes between the variables y and x. (A plot of y versus x, showing the general shape of the fitted model, could form an important part of your answer.)

Answer.

Visual

We start with a visual comparison of models (1) and (2). The dashed line marks $x = t$, where model (1) allows the mean to jump and model (2) does not. (The particular intercepts and slopes are arbitrary.)



Description

Splitting on the value of z_i, model (1) says

$$\mathrm{E}(y_i) = \begin{cases} \beta_0 + \beta_2 x_i & \text{if } x_i \le t, \\ (\beta_0 + \beta_1) + (\beta_2 + \beta_3) x_i & \text{if } x_i > t, \end{cases}$$

and model (2) says the same, with γ's in place of β's. So each model proposes a straight line on either side of t, with the following quantities of interest:

Quantity	Model (1)	Model (2)
Intercept for $x \le t$	β_0	γ_0
Slope for $x \le t$	β_2	γ_2
Intercept for $x > t$	$\beta_0 + \beta_1$	$\gamma_0 + \gamma_1$
Slope for $x > t$	$\beta_2 + \beta_3$	$\gamma_2 + \gamma_3$
Jump at $x = t$	$\beta_1 + \beta_3 t$	$\gamma_1 + \gamma_3 t = 0$

In model (1), the four β's are unconstrained, so the two lines are unrelated, and the mean may jump at $x = t$, by

$$\lim_{x \downarrow t} \mathrm{E}(y) - \mathrm{E}(y)\big|_{x=t} = \{(\beta_0 + \beta_1) + (\beta_2 + \beta_3)t\} - (\beta_0 + \beta_2 t) = \beta_1 + \beta_3 t.$$

In model (2), the constraint sets this jump to zero, so the two lines meet at $x = t$, at height $\gamma_0 + \gamma_2 t$, and the mean is continuous and piecewise linear. Substituting $\gamma_1 = -\gamma_3 t$,

$$\mathrm{E}(y_i) = \gamma_0 + \gamma_2 x_i + \gamma_3 (x_i - t) z_i, \qquad \text{(Q1 (a))}$$

so model (2) has three free mean parameters to model (1)'s four.

In both models, the y_i's are uncorrelated and vary about the mean with the same variance σ^2 on either side of t.

(b)

For each of the following two statements, decide whether the statement is correct. Justify your answers.

Statement 1: For model (1), the (ordinary) least squares estimator of β_2 depends only on those y_i's for which $x_i \le t$.

Statement 2: For model (2), the (constrained) least squares estimator of γ_2 depends only on those y_i's for which $x_i \le t$.

Answer. Statement 1 is correct. Statement 2 is incorrect.

Note: Both models share the $n \times 4$ model matrix X with rows $(1, z_i, x_i, x_i z_i)$. We utilize the following results:

- **Theorem 2.1**: Under Models II–VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $\underline{\lambda}' \in \mathcal{R}(X)$.
- **Theorem 2.8**: Under Models II–VII, the least squares estimate of an estimable $\underline{\lambda}'\underline{\beta}$ equals $\underline{r}' X' \underline{y}$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$.

Since all four β's are estimable, Theorem 2.1 puts every 1×4 unit vector in $\mathcal{R}(X)$, so X has full column rank. What's more, this means at least two distinct x_i lie on each side of t. Otherwise, the columns of X would be linearly dependent: if the $x_i > t$ all equaled some x_*, the fourth column would be x_* times the second; and if the $x_i \le t$ all equaled some x_*, the third column minus the fourth would be x_* times the first minus the second.

Statement 1

The two lines in model (1) share no parameters, so they are fit separately. To see this, let $\delta_0 \equiv \beta_0 + \beta_1$ and $\delta_1 \equiv \beta_2 + \beta_3$. Then the least squares objective function, $Q(\underline{\beta}) = (\underline{y} - X\underline{\beta})'(\underline{y} - X\underline{\beta})$, splits as

$$Q(\underline{\beta}) = \sum_{x_i \le t} (y_i - \beta_0 - \beta_2 x_i)^2 + \sum_{x_i > t} (y_i - \delta_0 - \delta_1 x_i)^2.$$

The map from $\underline{\beta}$ to $(\beta_0, \beta_2, \delta_0, \delta_1)$ is one-to-one and onto $\mathbb{R}^4$. So the first sum depends only on (β_0, β_2), the second only on (δ_0, δ_1), and the two pairs vary independently of one another. Hence Q is minimized by minimizing each sum separately. That is, $(\widehat{\beta}_0, \widehat{\beta}_2)$ minimizes the first sum, and $\widehat{\beta}_2$ is the least squares slope from regressing y on x using only the points with $x_i \le t$.

So $\widehat{\beta}_2$ depends only on the y_i's with $x_i \le t$, and Statement 1 is correct.

Statement 2

The key difference from Statement 1 is the constraint being imposed, "connecting" (geometrically speaking) the two lines together at $x = t$. Because of this constraint, the left-hand line must pass through the common value at $x = t$, and the y_i's with $x_i > t$ help locate that value.

Proof:

By (Q1 (a)), any $\underline{\gamma}$ satisfying $\gamma_1 + \gamma_3 t = 0$ has $X\underline{\gamma} = M\underline{\gamma}_*$, where $\underline{\gamma}_* \equiv (\gamma_0, \gamma_2, \gamma_3)'$ and M is the $n \times 3$ matrix with rows $(1, x_i, (x_i - t)z_i)$. As $\underline{\gamma}$ ranges over the constraint set, $\underline{\gamma}_*$ ranges over all of $\mathbb{R}^3$. So minimizing the objective function $Q(\underline{\gamma}) = (\underline{y} - X\underline{\gamma})'(\underline{y} - X\underline{\gamma})$ subject to the constraint is the same as minimizing $(\underline{y} - M\underline{\gamma}_*)'(\underline{y} - M\underline{\gamma}_*)$ with no constraint at all. That is, $\widehat{\gamma}_2$ is the ordinary least squares estimator of $\gamma_2 = \underline{\lambda}'\underline{\gamma}_*$, with $\underline{\lambda} \equiv (0, 1, 0)'$, under the model $\underline{y} = M\underline{\gamma}_* + \underline{\epsilon}$, which is of the form of Model II. This allows us to utilize known results (Theorem 2.8 as noted above).

What's more, γ_2 is estimable under this model, as M has full column rank. To see this, note that if $M\underline{\gamma}_* = \underline{0}_n$, then $X\underline{\gamma} = \underline{0}_n$, so $\underline{\gamma} = \underline{0}_4$ (by the Note above) and $\underline{\gamma}_* = \underline{0}_3$.

Applying Theorem 2.8 with M in place of X,

$$\widehat{\gamma}_2 = \underline{r}'M'\underline{y} = \underline{w}'\underline{y}, \qquad \text{where } \underline{w} \equiv M\underline{r} \text{ and } M'\underline{w} = M'M\underline{r} = \underline{\lambda}.$$

So $\widehat{\gamma}_2$ is a weighted sum of the y_i's, and it depends on y_i exactly when the weight w_i is nonzero. It then suffices to show that $w_i \neq 0$ for some i with $x_i > t$. To do so, we make use of two properties of the weights:

- Writing $\underline{w} = M\underline{r}$ entrywise, $w_i = f(x_i)$, where $f(x) \equiv r_1 + r_2 x + r_3 \max(x - t, 0)$. Like a model (2) mean, f is continuous and linear on each side of t.
- The first two rows of $M'\underline{w} = \underline{\lambda}$ give $\sum_i w_i = 0$ and $\sum_i x_i w_i = 1$.

With these properties in hand, suppose, for contradiction, that $w_i = 0$ for every i with $x_i > t$. Then f is linear on $[t, \infty)$ and zero at two distinct points there (again, making use of the above Note), so f vanishes on all of $[t, \infty)$. In particular, $f(t) = 0$. On $(-\infty, t]$, $f(x) = r_1 + r_2 x$ with $f(t) = 0$, so $f(x) = r_2(x - t)$ there, and

$$0 = \sum_{i=1}^{n} w_i = r_2 \sum_{x_i \le t} (x_i - t).$$

The last sum is negative, as its terms are nonpositive and not all zero (by the Note). So $r_2 = 0$, meaning every $w_i = 0$, which contradicts $\sum_i x_i w_i = 1$.

Hence $w_i \neq 0$ for at least one i with $x_i > t$. That is, $\widehat{\gamma}_2$ depends on some y_i with $x_i > t$, and Statement 2 is incorrect.

Q2

For the following questions 2 and 3, let $G = \begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix}$ be any generalized inverse of $\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}$ (the coefficient matrix of the constrained normal equations for Models IIa–VIIa).

Consider Models IIa–VIIa. Let $\widehat{\underline{\beta}}$ be the first component of any solution to the constrained normal equations.

Note: Throughout, A is $q \times p$ and $\underline{b}$ is $q \times 1$, the constraints $A\underline{\beta} = \underline{b}$ are consistent, and the constrained normal equations are

$$\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix} \begin{pmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{pmatrix} = \begin{pmatrix} X'\underline{y} \\ \underline{b} \end{pmatrix}.$$

G is partitioned to match the coefficient matrix, so G_{11} is $p \times p$, G_{12} is $p \times q$, G_{21} is $q \times p$, and G_{22} is $q \times q$.

(a)

Show that $X\underline{\beta}$ has a unique least squares estimator $X\widehat{\underline{\beta}} = XG_{11}X'\underline{y} + XG_{12}\underline{b}$ and that the vector of residuals is $\widehat{\underline{\epsilon}} = (I - XG_{11}X')\underline{y} - XG_{12}\underline{b}$.

Answer.

We utilize the following results:

- **Theorem 2.11**: Under Models IIa–VIIa, a linear function $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $\underline{\lambda}' \in \mathcal{R}\begin{pmatrix} X \\ A \end{pmatrix}$.
- **Theorem 2.12**: The constrained normal equations are consistent.
- **Theorem 2.13**: $Q(\underline{\beta}) \equiv (\underline{y} - X\underline{\beta})'(\underline{y} - X\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, if and only if the constrained normal equations have a solution whose first component is $\widehat{\underline{\beta}}$.
- **Corollary to Theorem 2.15**: $X\widehat{\underline{\beta}} = X\widetilde{\underline{\beta}}$ for the first components $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ of any two solutions to the constrained normal equations.
- **Theorem 1.2**: A matrix G is a generalized inverse of a matrix B if and only if $\underline{w} = G\underline{c}$ solves $B\underline{w} = \underline{c}$ for every vector $\underline{c}$ for which these equations are consistent.

To start, $X\underline{\beta}$ is estimable: each row of X is a row of $\begin{pmatrix} X \\ A \end{pmatrix}$, so each element of $X\underline{\beta}$ is estimable by Theorem 2.11. By definition, its least squares estimator is $X\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any minimizer of $Q(\underline{\beta})$ over the $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$. Said differently, by Theorem 2.13, $\widehat{\underline{\beta}}$ is the first component of any solution to the constrained normal equations. By the Corollary to Theorem 2.15, $X\widehat{\underline{\beta}}$ is the same for all such solutions, so the least squares estimator of $X\underline{\beta}$ is unique.

We must then find its form. By Theorem 2.12, the constrained normal equations are consistent, so by Theorem 1.2, one possible solution is

$$G\begin{pmatrix} X'\underline{y} \\ \underline{b} \end{pmatrix} = \begin{pmatrix} G_{11}X'\underline{y} + G_{12}\underline{b} \\ G_{21}X'\underline{y} + G_{22}\underline{b} \end{pmatrix}.$$

Its first component is $G_{11}X'\underline{y} + G_{12}\underline{b}$, so, by the uniqueness of $X\widehat{\underline{\beta}}$,

$$X\widehat{\underline{\beta}} = X(G_{11}X'\underline{y} + G_{12}\underline{b}) = XG_{11}X'\underline{y} + XG_{12}\underline{b},$$

whichever generalized inverse G is used. Hence the vector of residuals is

$$\widehat{\underline{\epsilon}} = \underline{y} - X\widehat{\underline{\beta}} = (I - XG_{11}X')\underline{y} - XG_{12}\underline{b}.$$

(b)

Use Theorem 2.14 and Lemma 2.13 to show (b) and (c).

$\mathrm{E}(\underline{\hat{\epsilon}}) = \underline{0}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

Answer.

We utilize the following results:

- **Theorem 2.14(ii)**: For any generalized inverse G of the coefficient matrix, $-A'G_{22}A = -A'G'_{22}A = X'XG'_{21}A = A'G_{21}X'X = X'XG_{12}A = A'G'_{12}X'X$.
- **Lemma 2.13**: For any generalized inverse G of the coefficient matrix, $X'X = X'XG_{11}X'X - A'G_{22}A$.
- **Lemma 1.3(ii)**: For any matrices M, P, and Q of appropriate dimensions, $M'MP = M'MQ$ implies $MP = MQ$.

To see the identity $XG_{11}X'X + XG_{12}A = X$, note that $X'XG_{12}A = -A'G_{22}A$ by Theorem 2.14(ii).

So by Lemma 2.13,

$$X'X(G_{11}X'X + G_{12}A) = X'XG_{11}X'X - A'G_{22}A = X'X = X'XI_p.$$

Then, noting Lemma 1.3(ii), with $M = X$, $P = G_{11}X'X + G_{12}A$, and $Q = I_p$, we have

$$XG_{11}X'X + XG_{12}A = X. \qquad \text{(Q2 (b))}$$

Then, recall that $\mathrm{E}(\underline{y}) = X\underline{\beta}$ under Models IIa–VIIa. So for any $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, part (a) and (Q2 (b)) give

$$\mathrm{E}(\underline{\hat{\epsilon}}) = (I - XG_{11}X')X\underline{\beta} - XG_{12}\underline{b} = (X - XG_{11}X'X - XG_{12}A)\underline{\beta} = \underline{0}_n,$$

where the second equality substitutes $\underline{b} = A\underline{\beta}$.

(c)

$XG_{11}X'$ and $I - XG_{11}X'$ are symmetric and idempotent matrices.

Answer.

We utilize Lemma 1.3(ii) (as noted in part (b): $M'MP = M'MQ$ implies $MP = MQ$), along with

- **Theorem 2.14(iv)**: For any generalized inverse G of the coefficient matrix, $X'XG_{11}X'XG_{11}X'X = X'XG_{11}X'X = X'XG'_{11}X'XG_{11}X'X = X'XG'_{11}X'X$.

We tackle the properties of symmetry and idempotence, in that order.

Symmetry of $XG_{11}X'$. By Theorem 2.14(iv), $X'X(G_{11}X'X) = X'X(G'_{11}X'X)$, so Lemma 1.3(ii) gives $XG_{11}X'X = XG'_{11}X'X$. Transposing, $X'X(G'_{11}X') = X'X(G_{11}X')$, and Lemma 1.3(ii) again gives

$$XG'_{11}X' = XG_{11}X', \qquad \text{that is,} \qquad (XG_{11}X')' = XG_{11}X'.$$

Idempotence of $XG_{11}X'$. By Theorem 2.14(iv), $X'X(G_{11}X'XG_{11}X'X) = X'X(G_{11}X'X)$, so Lemma 1.3(ii) gives

$$(XG_{11}X')(XG_{11}X')X = (XG_{11}X')X.$$

Transposing, and using the symmetry just shown, $X'X(G_{11}X'XG_{11}X') = X'X(G_{11}X')$. Lemma 1.3(ii) once more gives

$$(XG_{11}X')(XG_{11}X') = XG_{11}X'.$$

So $I - XG_{11}X'$ is symmetric, as I and $XG_{11}X'$ both are symmetric; and $I - XG_{11}X'$ is idempotent, as

$$(I - XG_{11}X')^2 = I - 2XG_{11}X' + (XG_{11}X')^2 = I - XG_{11}X'.$$

Q3

Take the model to be Model IIa and assume that $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ are both estimable. Let $\widehat{\underline{\beta}}$ be the first component of any solution to the constrained normal equations. Show that $\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'G_{11}\Lambda_2$.

Answer.

Note: As in Q2, let $G = \begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix}$ be any generalized inverse of $\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}$ (the coefficient matrix of the constrained normal equations for Models IIa–VIIa). Λ_1 and Λ_2 each have p rows, and $\Lambda_j'\underline{\beta}$ being estimable means that $\underline{\lambda}'\underline{\beta}$ is estimable for each column $\underline{\lambda}$ of Λ_j.

Additionally, we utilize the following results:

- **Theorem 2.16**: Under Models IIa–VIIa, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if the constrained conjugate normal equations, $X'X\underline{r}_1 + A'\underline{r}_2 = \underline{\lambda}$ and $A\underline{r}_1 = \underline{0}_q$, are consistent.
- **Theorem 2.17**: Under Models IIa–VIIa, the least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is given by $\underline{r}_1'X'\underline{y} + \underline{r}_2'\underline{b}$, where $\begin{pmatrix} \underline{r}_1 \\ \underline{r}_2 \end{pmatrix}$ is any solution to the constrained conjugate normal equations.
- **Theorem 1.2** (noted in Q2(a)): If $B\underline{w} = \underline{c}$ is consistent, then $\underline{w} = G\underline{c}$ solves it, for any generalized inverse G of B.
- **Corollary to Lemma 2.4**: For a random vector $\underline{x}$ with $\text{Var}(\underline{x}) = \Sigma$ and a matrix of constants L, $\text{Var}(L'\underline{x}) = L'\Sigma L$.

To start, let $\underline{\lambda}$ be any column of Λ_1 or of Λ_2. Since $\underline{\lambda}'\underline{\beta}$ is estimable, the constrained conjugate normal equations are consistent (Theorem 2.16). What's more, their coefficient matrix is the one for which G is a generalized inverse. So by Theorem 1.2, they are solved by

$$\begin{pmatrix} \underline{r}_1 \\ \underline{r}_2 \end{pmatrix} = G\begin{pmatrix} \underline{\lambda} \\ \underline{0}_q \end{pmatrix} = \begin{pmatrix} G_{11}\underline{\lambda} \\ G_{21}\underline{\lambda} \end{pmatrix}.$$

Collecting these solutions, column-by-column, let

$$R_1 \equiv G_{11}\Lambda_1, \qquad R_2 \equiv G_{21}\Lambda_1, \qquad S_1 \equiv G_{11}\Lambda_2, \qquad S_2 \equiv G_{21}\Lambda_2.$$

Then the constrained conjugate normal equations give

$$X'XR_1 + A'R_2 = \Lambda_1 \qquad \text{and} \qquad AS_1 = 0,$$

and Theorem 2.17, applied to each column, gives

$$\Lambda_1'\widehat{\underline{\beta}} = R_1'X'\underline{y} + R_2'\underline{b} \qquad \text{and} \qquad \Lambda_2'\widehat{\underline{\beta}} = S_1'X'\underline{y} + S_2'\underline{b}.$$

Next, note that under Model IIa, $\text{Var}(\underline{y}) = \sigma^2 I$, and $R_2'\underline{b}$ and $S_2'\underline{b}$ are constants, which do not affect the covariance. So by the Corollary to Lemma 2.4, with $L = (XR_1,\ XS_1)$,

$$\text{Var}\begin{pmatrix} \Lambda_1'\widehat{\underline{\beta}} \\ \Lambda_2'\widehat{\underline{\beta}} \end{pmatrix} = L'(\sigma^2 I)L = \sigma^2\begin{pmatrix} R_1'X'XR_1 & R_1'X'XS_1 \\ S_1'X'XR_1 & S_1'X'XS_1 \end{pmatrix},$$

whose upper-right block is $\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2 R_1'X'XS_1$.

Then, since $R_1'X'X = (\Lambda_1 - A'R_2)' = \Lambda_1' - R_2'A$ and $AS_1 = 0$, we may simplify this block further:

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2(\Lambda_1' - R_2'A)S_1 = \sigma^2\Lambda_1'S_1 = \sigma^2\Lambda_1'G_{11}\Lambda_2.$$

Q4

Prove Theorem 2.18, the generalization of the Gauss-Markov theorem to Models IIa–VIIa.

Answer.

The end goal is to prove the following Theorem, **Theorem 2.18:** Under Models IIa–VIIa, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is the first component of any solution to the constrained normal equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Under Model IIa, the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is uniformly less than that of any other linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, that is, $\underline{\lambda}'\widehat{\underline{\beta}}$ is the best (minimum variance) linear unbiased estimator (BLUE) of $\underline{\lambda}'\underline{\beta}$.

To prove this Theorem, we utilize Theorems 2.16 and 2.17 and the Corollary to Lemma 2.4 (all noted in Q3 above), along with

- **Lemma 2.9**: Under Models IIa–VIIa, a linear estimator $c + \underline{a}'\underline{y}$ of a linear function $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if there exists a vector $\underline{k}$ such that $c = \underline{k}'\underline{b}$ and $\underline{\lambda}' = \underline{a}'X + \underline{k}'A$.

Proof:

There are two claims here to prove: (1) that $\underline{\lambda}'\widehat{\underline{\beta}}$ is linear and unbiased under any of Models IIa–VIIa (ok, technically two things at once, but. . . y'know), and (2) that under Model IIa, $\underline{\lambda}'\widehat{\underline{\beta}}$ is the best linear unbiased estimator, i.e., it has the smallest variance among all linear unbiased estimators of $\underline{\lambda}'\underline{\beta}$.

Linear and unbiased: Since $\underline{\lambda}'\underline{\beta}$ is estimable, Theorem 2.16 gives vectors $\underline{r}_1$ and $\underline{r}_2$ with

$$X'X\underline{r}_1 + A'\underline{r}_2 = \underline{\lambda} \qquad \text{and} \qquad A\underline{r}_1 = \underline{0}_q.$$

Plugging these into Theorem 2.17,

$$\underline{\lambda}'\widehat{\underline{\beta}} = \underline{r}_1'X'\underline{y} + \underline{r}_2'\underline{b} = c + \underline{a}'\underline{y}, \qquad \text{where } c \equiv \underline{r}_2'\underline{b} \text{ and } \underline{a} \equiv X\underline{r}_1,$$

which is linear in $\underline{y}$. It is also unbiased, since $\underline{a}'X + \underline{r}_2'A = (X'X\underline{r}_1 + A'\underline{r}_2)' = \underline{\lambda}'$ is exactly the condition in Lemma 2.9 (with $\underline{k} = \underline{r}_2$).

Best: We compare two linear unbiased estimators of $\underline{\lambda}'\underline{\beta}$, one being $\underline{\lambda}'\widehat{\underline{\beta}} = c + \underline{a}'\underline{y}$, and any other, $c_0 + \underline{a}_0'\underline{y}$. By Lemma 2.9, $c_0 = \underline{k}'\underline{b}$ and $\underline{\lambda}' = \underline{a}_0'X + \underline{k}'A$ for some vector $\underline{k}$. Writing the difference in their weights as $\underline{d} \equiv \underline{a}_0 - \underline{a}$,

$$X'\underline{d} = X'\underline{a}_0 - X'X\underline{r}_1 = (\underline{\lambda} - A'\underline{k}) - (\underline{\lambda} - A'\underline{r}_2) = A'(\underline{r}_2 - \underline{k}),$$

so that

$$\underline{a}'\underline{d} = \underline{r}_1'X'\underline{d} = (A\underline{r}_1)'(\underline{r}_2 - \underline{k}) = 0.$$

Under Model IIa, $\mathrm{Var}(\underline{y}) = \sigma^2 I_n$, so by the Corollary to Lemma 2.4,

$$\mathrm{Var}(c_0 + \underline{a}_0'\underline{y}) = \sigma^2(\underline{a} + \underline{d})'(\underline{a} + \underline{d}) = \sigma^2\underline{a}'\underline{a} + \sigma^2\underline{d}'\underline{d} = \mathrm{Var}(\underline{\lambda}'\widehat{\underline{\beta}}) + \sigma^2\underline{d}'\underline{d}.$$

Since $\sigma^2 > 0$ and $\underline{d}'\underline{d} = 0$ iff $\underline{d} = \underline{0}_n$, the other estimator's variance is at least that of $\underline{\lambda}'\widehat{\underline{\beta}}$, with equality only when $\underline{a}_0 = \underline{a}$.

When $\underline{a}_0 = \underline{a}$, the two estimators are one and the same. In that case $(\underline{k} - \underline{r}_2)'A = (\underline{\lambda}' - \underline{a}'X) - (\underline{\lambda}' - \underline{a}'X) = \underline{0}_p'$. Since $A\underline{\beta} = \underline{b}$ is consistent, $\underline{b} = A\underline{\beta}_0$ for some $\underline{\beta}_0$, so $c_0 - c = (\underline{k} - \underline{r}_2)'A\underline{\beta}_0 = 0$.

Hence every linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$ other than $\underline{\lambda}'\widehat{\underline{\beta}}$ has strictly larger variance, whatever the value of $\sigma^2 > 0$, and $\underline{\lambda}'\widehat{\underline{\beta}}$ is the BLUE of $\underline{\lambda}'\underline{\beta}$, constraints (warts?) and all.

Q5

Suppose that $\underline{y}$ is an $n \times 1$ observable random vector that follows the model

$$\underline{y} = X\underline{\beta} + Z\underline{b} + \underline{e},$$

where X is an $n \times p$ known matrix of rank p^*, Z is an $n \times q$ known matrix, and $\underline{\beta}$ is a $p \times 1$ vector of unknown parameters. Furthermore, suppose that $\underline{b}$ and $\underline{e}$ are independent random vectors with $\underline{b} \sim N(\underline{0}, \sigma^2 D)$ and $\underline{e} \sim N(\underline{0}, \sigma^2 I)$, where σ^2 is an unknown positive parameter, and D is a given positive definite matrix.

(a)

Show that this model is an Aitken model (Model III) with H nonsingular.

Answer.

Note: In Model III, $\underline{y} = X\underline{\beta} + \underline{\epsilon}$ with $\mathrm{E}(\underline{\epsilon}) = \underline{0}_n$ and $\mathrm{Var}(\underline{\epsilon}) = \sigma^2 H$, where H is a given nonnegative definite matrix. So, writing $\underline{\epsilon} \equiv Z\underline{b} + \underline{e}$ (such that $\underline{y} = X\underline{\beta} + \underline{\epsilon}$), we need to show three things to conclude that this model is an Aitken model: (1) $\mathrm{E}(\underline{\epsilon}) = \underline{0}_n$; (2) $\mathrm{Var}(\underline{\epsilon}) = \sigma^2 H$ for a known, nonnegative definite H; and (3) that this H is nonsingular.

Since $\mathrm{E}(\underline{b}) = \underline{0}_q$ and $\mathrm{E}(\underline{e}) = \underline{0}_n$,

$$\mathrm{E}(\underline{\epsilon}) = Z\,\mathrm{E}(\underline{b}) + \mathrm{E}(\underline{e}) = \underline{0}_n.$$

For the variance, we utilize the Corollary to Lemma 2.4, which states that $\mathrm{Var}(L'\underline{x}) = L'\Sigma L$ for a random vector $\underline{x}$ with $\mathrm{Var}(\underline{x}) = \Sigma$ and a matrix of constants L. With $\underline{x} = \underline{b}$ and $L = Z'$, it gives $\mathrm{Var}(Z\underline{b}) = \sigma^2 ZDZ'$. Since $\underline{b}$ and $\underline{e}$ are independent, $Z\underline{b}$ and $\underline{e}$ are uncorrelated, so the variance of their sum is the sum of their variances:

$$\mathrm{Var}(\underline{\epsilon}) = \mathrm{Var}(Z\underline{b}) + \mathrm{Var}(\underline{e}) = \sigma^2 ZDZ' + \sigma^2 I_n = \sigma^2 H, \qquad \text{where } H \equiv I_n + ZDZ'.$$

H is known, since Z and D are known.

Next, H is nonnegative definite and nonsingular; both properties follow once we show that H is positive definite. To that end note that H is symmetric, since D is, and for any $n \times 1$ vector $\underline{v} \neq \underline{0}_n$,

$$\underline{v}'H\underline{v} = \underline{v}'\underline{v} + (Z'\underline{v})'D(Z'\underline{v}) \geq \underline{v}'\underline{v} > 0,$$

where the first inequality holds because D is positive definite, such that $(Z'\underline{v})'D(Z'\underline{v}) \geq 0$. So H is positive definite, and hence nonnegative definite. H is also nonsingular, since $\underline{v}'H\underline{v} > 0$ for every $\underline{v} \neq \underline{0}_n$ means $H\underline{v}$ can never be $\underline{0}_n$ unless $\underline{v} = \underline{0}_n$.

Combining all parts together, we have an Aitken model with $\underline{v}'H\underline{v} = \underline{v}'\underline{v} + (Z'\underline{v})'D(Z'\underline{v}) \geq \underline{v}'\underline{v}$ nonsingular.

(b)

Give a BLUE of any estimable function $\underline{\lambda}'\underline{\beta}$. (Simplify your results to the greatest extent possible.)

Answer. A BLUE of any estimable function $\underline{\lambda}'\underline{\beta}$ is

$$\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'(X'H^{-1}X)^{-}X'H^{-1}\underline{y}, \qquad \text{where } H^{-1} = I_n - Z(D^{-1} + Z'Z)^{-1}Z'.$$

To derive this result, we utilize the following results:

- **Theorem 2.19**: Under Model III (with H nonsingular), if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is any solution to the Aitken equations $X'H^{-1}X\widehat{\underline{\beta}} = X'H^{-1}\underline{y}$, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, and the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is uniformly less than that of any other linear unbiased estimator.
- **Lemma 2.15**: Corresponding to any $n \times n$ nonnegative definite matrix A of rank $r \geq 1$, there exists an $r \times n$ matrix S such that $A = S'S$. Moreover, for any $r \times n$ matrix S such that $A = S'S$, $\mathrm{rank}(S) = r$.

- **Theorem 2.3**: The normal equations $X'X\widehat{\underline{\beta}} = X'\underline{y}$ are consistent.
- **Theorem 1.2** (noted in Q2(a)): If $B\underline{w} = \underline{c}$ is consistent, then $\underline{w} = G\underline{c}$ solves it, for any generalized inverse G of B.

By part (a), the model is Model III with H nonsingular, so by Theorem 2.19, $\underline{\lambda}'\widehat{\underline{\beta}}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for any solution $\widehat{\underline{\beta}}$ to the Aitken equations. We then have two goals: (1) Write down a solution to the Aitken equations, and (2) simplify to the extent possible (simplify H^{-1}).

A solution to the Aitken equations: By part (a), H is $n \times n$, nonnegative definite, and of rank n, so Lemma 2.15 gives an $n \times n$ matrix R of rank n, that is, a nonsingular R, with $H = R'R$. Then $H^{-1} = R^{-1}(R^{-1})'$, and with $W \equiv (R^{-1})'X$ and $\underline{z} \equiv (R^{-1})'\underline{y}$,

$$W'W = X'R^{-1}(R^{-1})'X = X'H^{-1}X \qquad \text{and} \qquad W'\underline{z} = X'R^{-1}(R^{-1})'\underline{y} = X'H^{-1}\underline{y}.$$

So the Aitken equations are the normal equations $W'W\widehat{\underline{\beta}} = W'\underline{z}$, with W in place of X and $\underline{z}$ in place of $\underline{y}$. These are consistent by Theorem 2.3, so by Theorem 1.2, one possible solution is then given by

$$\widehat{\underline{\beta}} = (X'H^{-1}X)^{-}X'H^{-1}\underline{y}.$$

Simplifying H^{-1}: The goal here is to show that

$$H^{-1} = I_n - Z(D^{-1} + Z'Z)^{-1}Z'.$$

For the right-hand side, $D^{-1} + Z'Z$ must be nonsingular. Since D is positive definite, so is D^{-1}, as, for any $\underline{v} \neq \underline{0}_q$, $\underline{v}'D^{-1}\underline{v} = (D^{-1}\underline{v})'D(D^{-1}\underline{v}) > 0$.

So for any $\underline{v} \neq \underline{0}_q$,

$$\underline{v}'(D^{-1} + Z'Z)\underline{v} = \underline{v}'D^{-1}\underline{v} + (Z\underline{v})'(Z\underline{v}) > 0,$$

and $D^{-1} + Z'Z$ is positive definite. As with H in part (a), this means $(D^{-1} + Z'Z)\underline{v} \neq \underline{0}_q$ for every $\underline{v} \neq \underline{0}_q$, so $D^{-1} + Z'Z$ is nonsingular.

To verify the formula, multiply it by $H = I_n + ZDZ'$:

$$\begin{aligned}(I_n + ZDZ')\{I_n - Z(D^{-1} + Z'Z)^{-1}Z'\} &= I_n + ZDZ' - (Z + ZDZ'Z)(D^{-1} + Z'Z)^{-1}Z' \\ &= I_n + ZDZ' - ZD(D^{-1} + Z'Z)(D^{-1} + Z'Z)^{-1}Z' \\ &= I_n + ZDZ' - ZDZ' = I_n,\end{aligned}$$

where the second line factors $Z + ZDZ'Z = ZD(D^{-1} + Z'Z)$. Since H is a square matrix, a matrix that multiplies it to I_n is its inverse, which gives the formula for H^{-1}.

Substituting this H^{-1} into $\underline{\lambda}'\widehat{\underline{\beta}}$, the BLUE is

$$\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\{X'X - X'Z(D^{-1} + Z'Z)^{-1}Z'X\}^{-}\{X'\underline{y} - X'Z(D^{-1} + Z'Z)^{-1}Z'\underline{y}\}.$$

This form only requires inverting the $q \times q$ matrices D and $D^{-1} + Z'Z$, rather than the $n \times n$ matrix H.

(c)

Suppose that there exists a $p \times q$ matrix Q such that $Z = XQ$. Give a simple expression for a BLUE of any estimable function $\underline{\lambda}'\underline{\beta}$ and explain your reasoning. The expression that you give should NOT involve D.

Answer. A BLUE of an estimable $\underline{\lambda}'\underline{\beta}$ is its simple least squares estimator,

$$\underline{\lambda}'(X'X)^{-}X'\underline{y}.$$

We utilize Theorem 2.3 (the normal equations are consistent) and Theorem 1.2 (if $B\underline{w} = \underline{c}$ is consistent, then $G\underline{c}$ solves it, for any generalized inverse G of B), both noted in part (b), along with

- **Corollary 1 to Theorem 2.21**: Under Model III, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if there exists a matrix T such that $HX = XT$. (The notes call this matrix Q; we write T to keep it apart from the Q of this problem.)

By part (a), $H = I_n + ZDZ'$. Substituting $Z = XQ$, so that $Z' = Q'X'$, gives $H = I_n + XQDQ'X'$, and so

$$HX = X + XQDQ'X'X = X(I_p + QDQ'X'X) = XT, \qquad \text{where } T \equiv I_p + QDQ'X'X.$$

This is exactly the condition in Corollary 1 to Theorem 2.21, so the SLSE of every estimable $\underline{\lambda}'\underline{\beta}$ is a BLUE. The SLSE is $\underline{\lambda}'\widetilde{\underline{\beta}}$ for any solution $\widetilde{\underline{\beta}}$ to the normal equations $X'X\widetilde{\underline{\beta}} = X'\underline{y}$. These are consistent by Theorem 2.3, so by Theorem 1.2, one solution is $\widetilde{\underline{\beta}} = (X'X)^- X'\underline{y}$, which gives the BLUE

$$\underline{\lambda}'\widetilde{\underline{\beta}} = \underline{\lambda}'(X'X)^- X'\underline{y}.$$

And importantly, this expression does not involve D.

Q6

Consider the following model for a balanced two-way classification:

$$y_{ij} = \mu + \tau_i + \gamma_j + \epsilon_{ij} \quad (i = 1, \ldots, t, \;\; j = 1, \ldots, c);$$

where $\mu, \tau_1, \ldots, \tau_t, \gamma_1, \ldots, \gamma_c$ are unknown parameters; where $\epsilon_{11}, \ldots, \epsilon_{tc}$ are random errors, with $\mathrm{E}(\epsilon_{ij}) = 0$, $\mathrm{Var}(\epsilon_{ij}) = \sigma^2$, $\mathrm{corr}(\epsilon_{ij}, \epsilon_{ij'}) = \rho$ for $j' \neq j$, and $\mathrm{corr}(\epsilon_{ij}, \epsilon_{i'j'}) = 0$ for $i \neq i'$; and where σ^2 and ρ are unknown parameters satisfying $\sigma^2 > 0$ and $\rho > -1/(c-1)$ with $c > 1$. Establish whether or not the variance of the SLSE of every estimable function is uniformly less than or equal to that of any other linear unbiased estimator of that estimable function.

Hint: Writing $\underline{y} = (y_{11}, \ldots, y_{1c}, \ldots, y_{t1}, \ldots, y_{tc})'$, $\underline{\epsilon} = (\epsilon_{11}, \ldots, \epsilon_{1c}, \ldots, \epsilon_{t1}, \ldots, \epsilon_{tc})'$, and $\underline{\beta} = (\mu, \tau_1, \ldots, \tau_t, \gamma_1, \ldots, \gamma_c)'$, then the model matrix X (which is $(tc) \times (1 + t + c)$) and the variance matrix (which is $(tc) \times (tc)$) can be expressed as

$$X = \begin{pmatrix} \underline{1} & \underline{1} & \underline{0} & \cdots & \underline{0} & I \\ \underline{1} & \underline{0} & \underline{1} & \cdots & \underline{0} & I \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \underline{1} & \underline{0} & \underline{0} & \cdots & \underline{1} & I \end{pmatrix}. \qquad V = \mathrm{Var}(\underline{\epsilon}) = \sigma^2 \begin{pmatrix} A & & & \\ & A & & \\ & & \ddots & \\ & & & A \end{pmatrix}.$$

Above $\underline{1} = (1, \ldots, 1)'$ and $\underline{0} = (0, \ldots, 0)'$ are $c \times 1$ column vectors, while I be a $c \times c$ identity matrix and $A = (1 - \rho)I + \rho \underline{1}\,\underline{1}'$.

Answer. Yes, the variance of the SLSE of every estimable function is uniformly less than or equal to that of any other linear unbiased estimator of that estimable function.

To prove this, we utilize the following result:

- **Remark(s) following Corollary 2 to Theorem 2.21**: Under Model VII with variance V, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if there exists a Q such that $VX = XQ$, where the elements of Q may be functions of $\underline{\theta}$.

In Model VII, $\mathrm{E}(\underline{\epsilon}) = \underline{0}$ and $\mathrm{Var}(\underline{\epsilon}) = V$, where V may depend on unknown parameters $\underline{\theta}$. Here $\mathrm{E}(\underline{\epsilon}) = \underline{0}_{tc}$, and V depends on $\underline{\theta} = (\sigma^2, \rho)'$, so our model is a special case of Model VII. By the Remark(s), it then suffices to find a matrix Q, whose entries may depend on σ^2 and ρ, such that $VX = XQ$.

We find Q by computing VX directly from the block forms of X and V in the hint. Since V is block diagonal, with each diagonal block $\sigma^2 A$, each $\underline{1}$, $\underline{0}$, and I in X gets multiplied on the left by A:

$$VX = \sigma^2 \begin{pmatrix} A\underline{1} & A\underline{1} & \underline{0} & \cdots & \underline{0} & A \\ A\underline{1} & \underline{0} & A\underline{1} & \cdots & \underline{0} & A \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ A\underline{1} & \underline{0} & \underline{0} & \cdots & A\underline{1} & A \end{pmatrix},$$

using $A\underline{0} = \underline{0}$ and $AI = A$.

To simplify $A\underline{1}$, note that $\underline{1}$ is an eigenvector of A, with eigenvalue $a \equiv 1 + (c-1)\rho$. Since $\underline{1}'\underline{1} = c$,

$$A\underline{1} = (1 - \rho)\underline{1} + \rho\underline{1}(\underline{1}'\underline{1}) = \{1 + (c-1)\rho\}\underline{1} = a\underline{1}.$$

Substituting $A\underline{1} = a\underline{1}$ and $A = (1 - \rho)I + \rho\underline{1}\,\underline{1}'$,

$$VX = \sigma^2 \begin{pmatrix} a\underline{1} & a\underline{1} & \underline{0} & \cdots & \underline{0} & (1-\rho)I + \rho\underline{1}\,\underline{1}' \\ a\underline{1} & \underline{0} & a\underline{1} & \cdots & \underline{0} & (1-\rho)I + \rho\underline{1}\,\underline{1}' \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a\underline{1} & \underline{0} & \underline{0} & \cdots & a\underline{1} & (1-\rho)I + \rho\underline{1}\,\underline{1}' \end{pmatrix}.$$

Now compare the columns of VX with those of X; we'll see that every column of VX is a linear combination of the columns of X because:

- the first $1+t$ columns of VX (those with $a\underline{1}$ or $\underline{0}$ blocks) are $\sigma^2 a$ times the first $1+t$ columns of X;
- the last c columns of VX (those with $(1-\rho)I + \rho\underline{1}\,\underline{1}'$ blocks) are $\sigma^2(1-\rho)$ times the last c columns of X (the I blocks), plus $\sigma^2\rho$ times the first column of X (the $\underline{1}$ blocks), repeated c times (giving the $\underline{1}\,\underline{1}'$ blocks).

Collecting the coefficients of these combinations, column by column, into a matrix gives

$$VX = XQ, \qquad \text{where } Q \equiv \sigma^2 \begin{pmatrix} a & \underline{0}_t' & \rho\underline{1}' \\ \underline{0}_t & aI_t & 0 \\ \underline{0}_c & 0 & (1-\rho)I_c \end{pmatrix},$$

partitioned into blocks of sizes 1, t, and c to match the columns of X, with each 0 a block of zeros.

This Q exists for every $\sigma^2 > 0$ and $\rho > -1/(c-1)$. So, by the Remark(s) following Corollary 2 to Theorem 2.21 (uff what a statement), the SLSE of every estimable function is a BLUE, for every value of σ^2 and ρ.